

Test 13: Number variance ??(L)

Verdict: PASS | Generated: 2026-09-14 23:57:04 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

Long-range statistics: ??(L) ~ (1/??)lnL for GUE ? asymptotic curve comparison.

METHOD: Long-range statistics: ??(L) ~ (1/??)lnL for GUE ? asymptotic curve comparison.

Result

Test 13 [HARDCORE pass 2]: 4 sub-checks, 0 failed ? PASS

Plots - 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 13 --no-two-pass

Independent verification guide

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[23:57:04.365] [+615.394s] calc: ??(L=2.000) = 0.381460 (n_pos=50000, n_windows=3000, mean_count=2.00)

[23:57:04.366] [+615.394s] calc: ??(L=4.536) = 0.402894 (n_pos=50000, n_windows=3000, mean_count=4.54)

[23:57:04.367] [+615.395s] calc: ??(L=10.287) = 0.354897 (n_pos=50000, n_windows=3000, mean_count=10.30)

[23:57:04.368] [+615.396s] calc: ??(L=23.331) = 0.313517 (n_pos=50000, n_windows=3000, mean_count=23.34)

[23:57:04.369] [+615.397s] calc: ??(L=52.915) = 0.353407 (n_pos=50000, n_windows=3000, mean_count=52.91)

[23:57:04.369] [+615.398s] calc: ??(L=120.010) = 0.354226 (n_pos=50000, n_windows=3000, mean_count=120.02)

[23:57:04.370] [+615.398s] calc: ??(L=272.178) = 0.355610 (n_pos=50000, n_windows=3000, mean_count=272.16)

[23:57:04.371] [+615.399s] calc: ??(L=617.292) = 0.351103 (n_pos=50000, n_windows=3000, mean_count=617.29)

[23:57:04.372] [+615.400s] calc: ??(L=1400.000) = 0.342086 (n_pos=50000, n_windows=3000, mean_count=1399.99)

[23:57:04.378] [+615.407s] calc: ??(L=2.000) = 0.417134 (n_pos=8401, n_windows=3000, mean_count=2.00)

[23:57:04.379] [+615.407s] calc: ??(L=4.536) = 0.484449 (n_pos=8401, n_windows=3000, mean_count=4.53)

[23:57:04.380] [+615.408s] calc: ??(L=10.287) = 0.609118 (n_pos=8401, n_windows=3000, mean_count=10.29)

[23:57:04.380] [+615.408s] calc: ??(L=23.331) = 0.691228 (n_pos=8401, n_windows=3000, mean_count=23.33)

[23:57:04.381] [+615.409s] calc: ??(L=52.915) = 0.745288 (n_pos=8401, n_windows=3000, mean_count=52.92)

[23:57:04.381] [+615.409s] calc: ??(L=120.010) = 0.783139 (n_pos=8401, n_windows=3000, mean_count=120.02)

[23:57:04.382] [+615.410s] calc: ??(L=272.178) = 0.897044 (n_pos=8401, n_windows=3000, mean_count=272.18)

[23:57:04.382] [+615.410s] calc: ??(L=617.292) = 0.908534 (n_pos=8401, n_windows=3000, mean_count=617.29)

[23:57:04.383] [+615.411s] calc: ??(L=1400.000) = 0.811269 (n_pos=8401, n_windows=3000, mean_count=1400.00)

[23:57:04.385] [+615.413s] calc: ??(L=23.331) = 0.324730 (n_pos=50000, n_windows=3000, mean_count=23.33)

[23:57:04.386] [+615.414s] calc: ??(L=23.331) = 0.306048 (n_pos=50000, n_windows=3000, mean_count=23.34)

[23:57:04.387] [+615.415s] calc: ??(L=23.331) = 0.318012 (n_pos=50000, n_windows=3000, mean_count=23.32)

CONSOLE CAPTURE

??

TEST 13 (secondary, HARDCORE): Objection 2 (extended): Advanced RMT Diagnostics ? ?? rigidity audit

??

L	??_data	Poisson=L	data/Pois
2.0	0.3815	2.0000	0.1907
4.5	0.4029	4.5359	0.0888
10.3	0.3549	10.2874	0.0345
23.3	0.3135	23.3315	0.0134
52.9	0.3534	52.9150	0.0067
120.0	0.3542	120.0097	0.0030
272.2	0.3556	272.1783	0.0013
617.3	0.3511	617.2922	0.0006
1400.0	0.3421	1400.0000	0.0002

N2 monotone : 0 dips >25% across 8 grid steps (limit 1)

N3 GUE band : 9/9 finite pairs have data/GUE ? [0.2, 5] (limit 80%)

N4 seed noise: ??(L=23.3) over 3 seeds: 0.3247/0.3060/0.3180, spread=5.9% (limit 40%)

??

- N1 beat Poisson PASS ? 9/9 L-values
- N2 monotone PASS ? 0 dips
- N3 GUE band PASS ? 9/9 in band
- N4 seed noise PASS ? spread = 5.9%

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